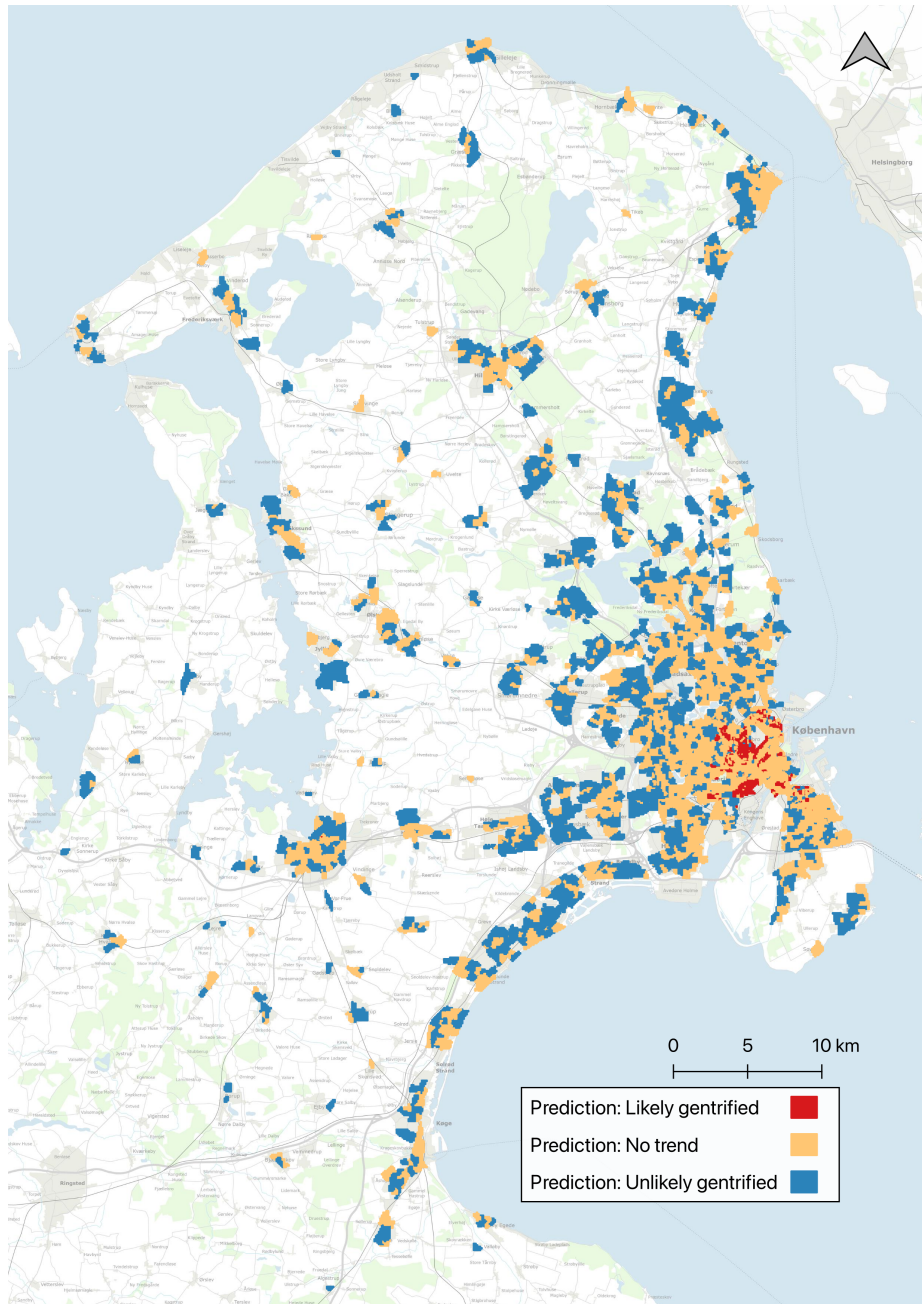


Predicting Gentrification with Machine Learning

Surveying, Planning and Land Management
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Abstract:

Gentrification is a growing topic in the big cities that have seen a consistent population growth through the years. The ambiguous nature of gentrification and its consequences is derived from past research but it is unclear how it should be viewed in a Danish context. This study aims to predict areas prone to gentrification which is done by defining gentrification and its driving forces based on literature.

This knowledge is used to design a XG-Boost machine learning model based on nationwide socioeconomic data from The Danish Neighborhood Atlas. The final model predicts gentrified areas in 2020 with an overall accuracy of 79%. These predictions point to gentrification only occurring in bigger cities, where displacement is likely. The findings can provide city planners and policymakers with a tool to mitigate or counteract gentrification with displacement as a consequence. This can help make the big cities more equitable especially for marginalized and vulnerable communities.

The content of this report is freely available, but publication (with reference) may only be pursued due to agreement with the author.

Synopsis in Danish

I takt med øget urbanisering i verdens storbyer har man set en markant stigning i efterspørgslen på boliger. Byernes attraktivitet forårsager mærkbare værdistigninger, hvilket kan føre til gentrificering. Gentrificering som term beskriver en udskiftning i befolkningssammensætning, som i praksis betyder, at den lavere sociale klasse bliver "presset" ud ad bydele til fordel for den højere sociale klasse. Denne tendens kan dokumenteres og kortlægges ud fra blandt andet socioøkonomiske befolkningsdata, hvilket er velbeskrevet i amerikansk, engelsk og australsk kontekst. I Danmark er der dog et videnshul som er værd at undersøge nærmere, hvilket har ført til udarbejdelsen af en problemformulering bestående af følgende arbejdsspørgsmål:

Hvordan kan gentrificering forstås i en dansk kontekst og er gentrificering det målbart?

I hvilken grad kan en machine learning model baseret på spatiale modeller, forudsige gentrificering i danske byer og bruges i en byplanlægningsproces?

Denne indledende undren besvares ud fra et litteraturstudie, hvor termet gentrificering og dets konsekvenser udspecificeres med udgangspunkt i eksisterende forskning. Dette gøres ved at se på fremtrædende definitioner, hvilke forudsætninger der fremmer gentrificering samt hvordan dette kan placeres i en dansk kontekst. Det undersøges også hvordan målbare socioøkonomiske data kan bruges til at identificere gentrificerede områder via machine learning (ML). Med henblik på at udvikle en ML model som kan anvendes i dansk kontekst, ses der nærmere på et udvalgt datasæt fra NabolagsAtlas fra perioden 1990-2020. Datasættet bygger på stedfæstet data fra Danmarks Statistik og dækker over Københavns Kommune, Frederiksberg Kommune og de resterende kommuner der fremgår indenfor Fingerplanens afgrænsning. På baggrund af gentrificeringstendenser bevist på Vesterbro og Nørrebro bliver data bearbejdet i ML modellen til at udpege områder med lignende mønstre. Udformningen af modellen baseres på XGBoost og yderligere optimering resulterer i en nøjagtighed på 79% til at identificere områder, der er tilbøjelige til at være gentrificeret.

Denne proces og dens resultater kortlægges og modellens anvendelighed diskuteres ud fra et planlægningsperspektiv. Her vurderes det, at modellen er brugbar som præventivt værktøj til byplanlæggere for at hæmme gentrificering og konsekvenserne derved. Slutvist konkluderes det, at ML modellen og dens udarbejdelse imødekommer problemformuleringens formål og der reflekteres yderligere over ML modellens forbedringspotentiale.

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PREFACE

This semester project was prepared by four students as part of the 1st semester of the Master's degree programme in Surveying, Planning and Land Management at Aalborg University Copenhagen in the Autumn of 2025.

The group would like to thank supervisors Jamal Jokar and Janni Sørensen for their valuable and professional guidance and Elise Stenholt Lange for providing us with data from the Danish Neighborhood Atlas developed by AAU BUILD.

Reader's Guide

The project report has been prepared with the intention of being printed double-sided. References are cited according to the Chicago style. In the text, the author, year of publication, and page number are indicated ([Author, year, page number]). Figures and images that do not have a literature reference in the project report are produced by the project group. Essential appendices are placed in the Appendix, which appears at the end of the project report. References in the digital version of the report are inserted as hyperlinks, allowing the reader to click on, for example, figure numbers, appendix letters, or page numbers in the table of contents to navigate directly to the selected figure, section, or page.

The codebase can be viewed via the GitHub repository:

https://github.com/jonashansen886-cyber/Gentrification_model_Copenhagen

In the repository some of the raw data files are replaced by dummy files to comply with guidelines from the Danish Neighborhood Atlas. The raw data is in the attached zip file and can manually be replaced as described in the README file in the repository.

PART I

INTRODUCTION

1 | Background and Motivation

In 2016 Jan Christensen, who is the former Chief City Architect of Copenhagen in the period 2001 to 2010 was quoted as saying “Copenhagen is going to be a city for the rich, there is not much we can do about that, really”. This age was in the wake of initiatives like the Danish Urban Renewal Act in Vesterbro and saw the establishment of several culture institutions, the Copenhagen Metro Line in 2002 as well as the first building in the newly established Ørestaden (Andersen, 1998). With 1,000 newcomers to the capital every month around 2016, Christensen predicted a great stress on the Copenhagen housing market, stating, “It is a global demographic trend that the highly educated and the rich will want to live in the big cities” (Kjeldtoft, 2016).

In present day it is evident that Jan Christensen was right. The Copenhagen municipal election of 2025 was an election about housing. A significant increase in population in the last 10 years alongside rising house prices and rent prices has made the housing market a madhouse for ordinary citizens. The average paid rent of a 60 m² private rented apartment in Copenhagen is DKK 7,900, while the average rent of available apartments of equal size is DKK 12,000 (a difference of $\approx 52\%$). Meanwhile, public housing of equal size costs DKK 5,100 (Københavns Kommune, 2025).

Public housing in Denmark has generally been in decline and saw its peek back in the 1970’s. It was the base for solving the housing shortage in the post war years and Denmark saw approximately 10,000 residences being built every year towards the 1990’s (Kolstad & Larsen, 2025). Copenhagen also saw this trend with approximately 1,000 public housing residences built per year, forming the majority of the building industry. From 2010-2016, newly built public housing only contributed to $\approx 11\%$ of the cumulative housing stock, which is the lowest of the share compared to the three second largest cities in the same time period (Københavns Kommune, 2018).

In the last 10 years there has only been built 5,100 public housing residences in Copenhagen and present legislation strives to focus the city’s supply towards the marginalized population (Københavns Kommune, 2025). This combination constitutes a framework that limits the livability for the working class. As scholars states, the lack of development in public housing can contribute to trends of gentrification, as the housing stock of private ownership and private renting will increase equivalently (Cohen & Pettit, 2019).

The present Danish Government has prioritized to advance the constrained framework of developing public housing to accommodate a housing market under pressure and prevent the prediction, that Jan Christensen made back in 2016 (Birch & Tofte, 2025).

These trends that Copenhagen sees can be interpreted as signs of gentrification, which is the result of a development in demographic, economic, residential and business structure. It is the idea of the city culture attracting the financially stable population and the physical environment transforming, to conform to the middle and upper class at the expense of the lower class population. It is also the replacing of authentic pubs and local shops with coffee shops, wine bars and business chains, which is exemplified by the correlation of every Starbucks coffee shop in NYC neighborhoods linking to an increase in surrounding house prices by 0.5% (Glaeser et al., 2018).

Locals who are responsible of business life in the district of Sydhavnen in Copenhagen, has taken it into their own hands to avoid these tendencies. When applying for a business it is paramount that you have local anchoring and a unique identity to ensure an authentic business life. As the chief consultant Lars Strand elaborates:

“It is very important that Sydhavnen got its own identity. We do not want to have copies of the same shops that exist in Vesterbro and Valby. It should be shops and businesses that buy in on the neighborhood” (Henriksen, 2025).

The concept of gentrification is an advanced topic but is important to understand in the context of large cities like Copenhagen. As mentioned the gentrification process can have consequences for the lower class citizens and can have a long-term impact on otherwise affordable neighborhoods. In accordance with the United Nations Sustainable Development Goals (SDG) number 11 called "Sustainable cities and communities", it is important to ensure both affordable housing and inclusive environments in the rapidly growing cities. This is elaborated in sub-goal 11.1, which states: "(...) ensure access for all to adequate, safe and affordable housing and basic services and upgrade slums" (The Global Goals, n.d.). By analyzing gentrification, its consequences and shedding light on possible vulnerable areas it is also consistent with working towards goal number 10 "Reduced inequalities" and goal number 16 "Peace, justice and strong institutions" (United Nations, n.d.).

2 | Problem Statement and Research Questions

Currently, there is no clear answer to which areas in Denmark that are prone to gentrification. There is a lack of quantitative research and a need for city planners to make considerate decisions from a broad socioeconomic dataset. Similar studies in other countries have been made using machine learning models to determine and map trends of gentrification.

To investigate gentrification in Danish cities, it is first necessary to define gentrification, as well as which areas have been exposed to gentrification. In addition, the gentrification process must be investigated to find out what drives gentrification but also find parameters on which it can be measured. These considerations lead to the following research questions:

How can the concept of gentrification be interpreted in a Danish context and is it measurable?

To what extent can machine learning based spatial models predict neighborhoods at risk of gentrification in Danish cities, and how can these predictions inform urban planning decisions?

3 | Literature Review

This chapter will be a systematically structured review of the existing knowledge from different literature about gentrification and machine learning (ML). Literature is compared, evaluated and organized appropriately in relation to the specific topics. The review is meant to define gentrification, highlight measurements and explore how gentrification can be predicted in a Danish context.

Gentrification is a broad subject with lots of different perceptions and inconsistencies on how the term is defined. The inconsistencies stems from the highly subjective experiences regarding the positive or negative consequences of gentrification. In 1964 Ruth Glass first defined gentrification, as the process where the working class inhabitants of London were moved out, so the upper middle class could move into newly renovated large houses. This creates a competition for space, where the ones who are less well off gets displaced to less desirable living quarters (Glass, 1964).

The topic of gentrification has gained increased attention over time. This direction is clear when looking at the number of materials concerning gentrification in the database for peer-reviewed research literature, Scopus. The graph in figure 3.1 shows the number of published literature which include the word *gentrification* in the title, abstract or keywords. The search is limited to only english publications which leads to 6,931 search results. It shows a significant focus on gentrification and a rise during the last decade. The search was conducted on the 4th of November 2025 and therefore the number of articles from 2025 are expected to increase.

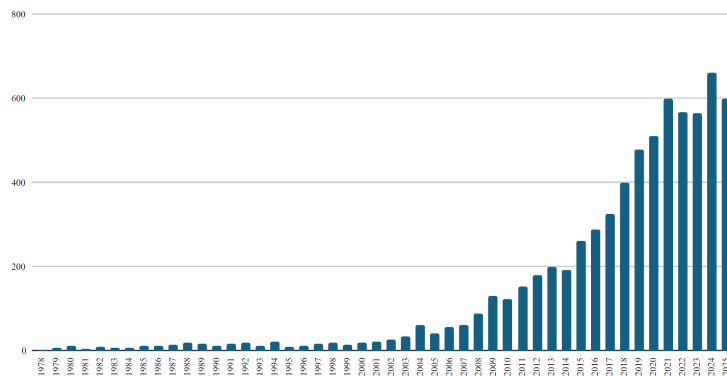


Figure 3.1: The number of peer-reviewed literature in Scopus regarding gentrification in relation to the year of publication.

3.1 Defining Gentrification

This project will focus on gentrification in a Danish context. It is therefore relevant to compare how the definition differ from country to country. The Danish Dictionary defines gentrification as: “The transformation of an urban area characterized by gradually rising housing prices and a changing population composition as a result of in- and out-migration, urban renewal, the merging of small apartments, changes in housing types, or similar developments” (The Danish Dictionary, n.d.). The Danish definition has a mostly neutral outlook on gentrification that does not mention consequences. Even more so is the Oxford Dictionary definition that describes gentrification as: “The process of renovating and improving a house or a district so that it conforms to middle class taste” (Knorr, 2019, p. 3) This definition draws on the same principles as the definition from Ruth Glass where the focus is tailored to middle class citizens. Both definitions focuses on how city planning can accommodate the middle class citizens, but the lack of consequences mentioned can be interpreted as a positive view. Ruth Glass includes displacement as a mean to make room for the middle class, whereas the Oxford definition does not include how it aims to conform to the middle class. These differences in how gentrification is defined in dictionaries can have an impact on how the term is perceived by the general public (Drew, 2018).

Researchers have defined the term, which has varied according to the local context and the purpose of the study. The National Center for Smart Growth at the University of Maryland defines gentrification for a specific place in the region as: “An increase in home prices and the population’s education level that was greater than the increases that occurred in the region as a whole.” (Cohen & Pettit, 2019, p. 5). This definition relies on the rise of housing prices and education level. Furthermore slowly evolving gentrification may be overlooked with this definition.

The Institute on Metropolitan Opportunity at the University of Minnesota Law School choose to define gentrification with more fluidity in the view of different measurements: “Displacement of lower-income households by higher-income residents, replacement and/or rehabilitation of housing stock, and displacement of racial minorities by higher-income white residents.” (Institute on Metropolitan Opportunity, 2016, p. 3). This definition includes types of displacement as part of gentrification and covers a broader range of neighborhood changes that are reliant on racial and economic factors. This definition is expressed similarly in the work by Zuk et al. (2015). In this article the American conditions regarding race and ethnicity is explored and viewed as a central part of neighborhood formation and transformation. This is due to the historic factors of integration and race-segregation that has formed America and its social classes (Zuk et al., 2015, p.7).

For this report the definition from The Danish Dictionary (The Danish Dictionary, n.d.) will be used as a starting point. Most definitions found in the literature focus on aspects that match the available data used in their respective studies. For example, Cohen and Pettit (2019) focuses only on education level and home prices. For our purposes, this is insufficient, as the data available includes many more variables that can help in defining gentrification in Greater Copenhagen and Denmark as a whole (Lange et al., 2023). The available data from DNA will be further elaborated in chapter 7 on page 29.

3.1.1 Additional Key Terms Concerning Gentrification

When defining gentrification it is evident that it is a process and a product of existing socioeconomic, political and demographic change. In addition to gentrification it is relevant to clarify other terms that are relevant to understanding gentrification in this study.

Neighborhood Change and Neighborhood Revitalization

Neighborhood change covers the general change and development in economic, social and spatial factors, which influence both the environment and its residents. Whereas the change can be interpreted as positive or negative, the term neighborhood revitalization focuses on the positive outcome of changing the environment with either public or private investment e.g. renovations or conversion of public space (Zuk et al., 2015, pp. 11–15).

Residential Mobility and Residential Displacement

Residential mobility is the general movement of residents between environments. This can be based on cultural, class-related, economic or racial forces that attract residents and is regarded as the voluntary act of movement. This is in contrast to residential displacement, where the same factors can result in repelling residents involuntarily based on e.g. higher rents, change in population composition or eviction (Zuk et al., 2015, pp. 5–7).

Social Classes

The social classes are often used to describe gentrification in the literature. Thus these terms are clarified further. The social classes are defined in a Danish context by The Economic Council of the Labour Movement¹ who uses information about education, income and position on the labor market. The population is divided in five groups: upper class, upper middle class, middle class, working class and long-term unemployed.

¹In danish: Arbejderbevægelsens Erhvervsråd

Roughly speaking gentrification occur when lower income households like the working class and the long-term unemployed are replaced with higher income residents like the middle and upper class (Institute on Metropolitan Opportunity, 2016, p. 3 & Smith, 1982, p. 139).

The middle class is defined as people who have an annual income of less than DKK 785,000 before tax (according to price level in 2018) and are either wage earners with management responsibilities, self-employed or they have a short or medium length tertiary education. The working class is defined as people with only a secondary education, vocational education or no education (Arbejderbevægelsens Erhvervsråd, 2020, p. 8). With these definitions it is worth noting that the gentrifying group does not have to be particularly wealthy but can belong to the middle class.

Scale of Neighborhoods

For the purpose of identifying neighborhoods as gentrified it is important to distinguish whether or not the given neighborhood is too small or too big to be considered gentrified. George Galster (2001) defines in his paper *On the Nature of Neighborhood* the neighborhood as: "... the bundle of spatially based attributes associated with clusters of residences, sometimes in conjunction with other land uses" (Galster, 2001). This definition is then applied to Suttles (1972) who classifies it into four distinct sizes based on what the inhabitants in the neighborhood determine.

1. Block face - the close proximity, where children are permitted to play without supervision - one to two streets.
2. Defended neighborhood – a small local area with a common identity.
3. Community of limited liability – a larger area, commonly consisted of local governmental district
4. Expanded community – an entire sector of the city.

Galster argues that neighborhoods cannot be defined with a single, unambiguous size. Instead they consist of a bundle of spatially based attributes that together create 'neighborhoodness'. These domains include, but are not limited to: Urban morphology, mobility and utility infrastructures, demographic composition, class status characteristics, environmental conditions, and proximity to recreational activities (Galster, 2001).

3.2 Gentrification in a Danish Context

To better understand gentrification in a Danish context, the same review as on figure 3.1 was conducted by adding 'Copenhagen' to the search. This produces 26 results in the period from 1994 to 2025. This search was conducted the 4th of November 2025 and the quantity is presented on figure 3.2 below. Comparing figure 3.1 and 3.2 shows that the increasing international focus on gentrification has not quite reached Denmark.

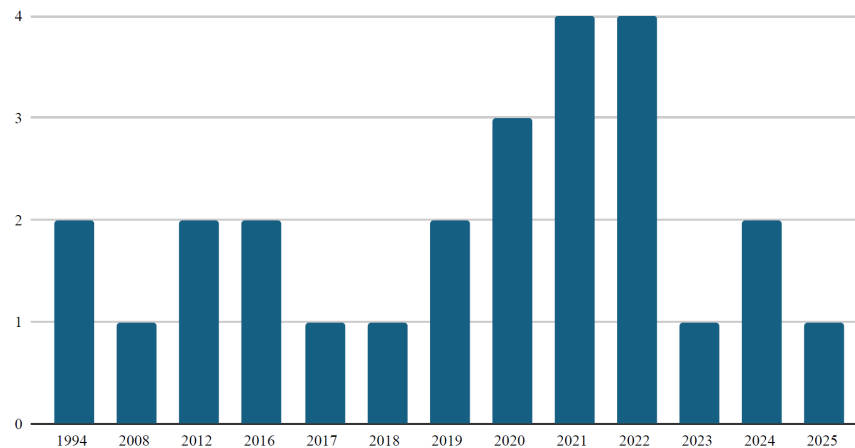


Figure 3.2: The number of peer-reviewed literature in Scopus regarding both Copenhagen and gentrification in relation to the year of publication

Andersen (1998) compares the socioeconomic change in deteriorated residences throughout cities in Denmark with the development in similar kinds of residences that have been renewed by public supported programs. The paper follows up on the Danish Urban Renewal Act undertaken in the 1990's. Indicating that deteriorated housing attracts low income and unemployed citizens and the younger age group. When public supported housing plans are introduced, the number of high income residents increase but this change was not considered in 1998 (Andersen, 1998). The date of publication is worth taking into account because the study may show a low degree of gentrification in Denmark due to its publication date of 1998.

Larsen and Hansen (2008) sheds a focused light on the situation of gentrification in Copenhagen. Their research laid the foundation for similar research in a Nordic context. In the article the intentions of the urban renewal of Inner Vesterbro in Copenhagen in the 90's is contrasted to the actual outcome. The intentions of renewal was to upgrade the shabby buildings with consideration to the original residents, however the result of the renewal was not as intended. The municipality had experiences with conflicts from the top-down renewal of Nørrebro in the 70's and the lessons from those projects influenced the more

inclusive approach in Vesterbro with a greater extent of public participation (Larsen & Hansen, 2008, p. 2436). Despite the well-meaning intentions the article points out that the urban renewal caused a change in the population from socioeconomically vulnerable inhabitants to middle-class inhabitants. It is argued that the reason behind is ambiguous policies among other things. The case around Inner Vesterbro reveals how urban renewal policies are framed as socially inclusive but have driven state supported gentrification.

The article draws the conclusion that gentrification has occurred at Vesterbro based on statistics on types of property owners and the socioeconomic composition of the inhabitants. The statistics shows among other things that the property ownership shifted in the period 1997-2005 from being dominated by private rental to private housing cooperatives² (Larsen and Hansen, 2008, p. 2443, Vidal, 2019, p. 162). Furthermore in the same period the percentage of wage earners increased and the number of unemployed decreased (Larsen & Hansen, 2008, p. 2441). The article also points out that a part of the displaced inhabitants of Vesterbro moved to the neighboring area of Sydhavn, where more affordable housing was accessible (Larsen & Hansen, 2008, p. 2443). The city of Copenhagen attempted to push out the marginalized and vulnerable inhabitants of Vesterbro, to make room for the middle class.

As Hansen et al. (2001) cites: “We have made a study that shows that we get all the trash, pardon the expression, because of the cheap housing. The middle class, who we would want to stay, moves to Sweden. (Holger Bisgaard, Head of Planning, Copenhagen, 26. January 2000)” (Hansen et al., 2001, p. 862). A part of this displacement-strategy was to renovate and change the apartments, which were predominantly leasehold, to private housing cooperatives. As part of funding this renewal the rent increased by around 50%, which in turn forced out the groups who could not afford this new high rent (Larsen & Hansen, 2008; Vidal, 2019).

The case of gentrification of Inner Vesterbro is further elaborated in Vidal (2019) with a case study on the effects of collaborative housing in Vesterbro after the renewal. It shows that the effects of renovating the rental housing and transforming it to private owned collaborative housing caused the original residents to move from the area in favor of residents with higher income and buying power (Vidal, 2019). This was further emphasized in 2004 with the ability to now use the housing cooperative shares as collateral for personal credit (Vidal, 2019, p. 168). Bredahl et al. (2021) also argues that the biggest signs of gentrification in Copenhagen are seen in especially the neighborhood Vesterbro, but also in The city centre and Amager West, when looking at change in income and

²In Danish: Private andelsboligforeninger

educational level. However, it should be considered that the source only uses these two criteria to measure gentrification (Bredahl et al., 2021, pp. 56–63). Whereas most other studies examine a least 3-4 criteria (e.g. property value, occupation, family composition) (Cohen & Pettit, 2019; Knorr, 2019; Reades et al., 2019; Vidal, 2019; Yoo, 2023).

As described in previous section 3.1 race and ethnicity is an important factor for gentrification to occur in an American context. Even though the impact is bigger in the US, it is not without importance in a context of Copenhagen. Loftkjær and Wredstrøm (2022) looks into how Nørrebro is underway to become gentrified based on the increasing average income and housing prices but also the dropping percentage of residents being immigrants of non-Western origin. It is estimated that from 2009 to 2015 the percentage of residents of Nørrebro being immigrants and descendants from non-western countries decreased from 27% to 18.4% (Loftkjær & Wredstrøm, 2022, p. 5). Due to the lack of literature and data on the progression of racial and ethnic composition in Copenhagen through time, this factor will not be part of our analysis.

The amount of literature about gentrification in Denmark is limited and the data used is not completely up to date. It is still applicable though, including the consensus about Inner Vesterbro being the most gentrified area in Copenhagen. No sources have been found regarding models that identify areas in larger parts of Denmark, that are prone to gentrification. The subject of predicting gentrified areas are therefore yet to be covered in a Danish context.

3.3 Drivers and Indicators of Gentrification

There are different factors indicating that gentrification in a given neighborhood is taking place. Some factors drive gentrification and other factors are an outcome of gentrification e.g. decreasing unemployment rate or increasing housing prices. Larsen and Hansen (2008) points out several driving factors: Policies about housing, urban renewals and commodification of space. Some external factors can be policies regarding interest rate and tax deduction, that can create beneficial conditions for investment.

Furthermore, when the government or private investors plan and execute urban renewals the interest in the area is raised. The renewal can consist of upgrading the existing housing, public spaces and public transportation. These driving forces cannot be directly quantified, but other factors that are indicators of the driving forces are measurable.

A wide range of data is used among the literature to quantify and measure gentrification. Two of the most popular measurement parameters are income and educational level, where a rise is seen as a sign of gentrification. Larsen and Hansen (2008) furthermore inspect how the unemployment rates and the property ownership have changed in Inner Vesterbro, where the unemployment rate has drastically dropped while the housing ownership has shifted from private rental to owner occupied housing (Larsen & Hansen, 2008, p. 2443).

Another perspective is presented by Cohen and Pettit (2019), who point out resident characteristics like age, race, income and education as the baseline after which more factors can be added like housing prices, vacant and blighted properties or crime rates. The types of indicators are many and it all depends on the context and what measurements are most relevant. Often the limiting factor is the available data. For example, rental prices are not publicly available in Denmark although it would be relevant to investigate. On the contrary race or ethnicity are less relevant in a Danish context.

When analyzing the numbers it is important to compare with the national or municipal average to make sure any changes are extraordinary and not just following the broader tendency (Cohen & Pettit, 2019, p. 8). For example the disposable income in Denmark has increased with 57% from 1990 to 2020, so the local rise has to exceed 57% to stand out (Lange et al., 2023). Furthermore the measurements should be compared to the driving factors like urban renewal policies but also social movements like global crises to spot if they are codependent.

3.4 Predicting Gentrification

As the previous section has explored how the definitions of gentrification can be quantified, this section means to put the quantities of data in context of a prediction model. Prediction models for neighborhood change have existed since the 1980's with the goal of monitoring housing conditions and map out indicators of declining settlements (Chappel & Zuk, 2016, p. 112). These 'early-warning systems' paved the way for similar models, which made it possible for municipalities, cities and other stakeholders to form an overview of demographic trends to figure out where resources should be spent on development (Reades et al., 2019, p. 925).

As the public sector have gotten hold of more refined and reliable data, these models have been expanding through time and have also been used to predict gentrification. This is mostly evident in the US, where planners in the past were focused on mapping out

underdeveloped areas to insure development (Chappel & Zuk, 2016, p. 110). However present prediction models are focusing on insuring a sustainable and cautious neighborhood development and mitigating displacement (Chappel & Zuk, 2016, p. 120).

Present prediction models have made use of machine learning (ML) to utilize data efficiently and find specific patterns within. ML is a way to model large datasets, find correlations in the data and use these correlations to form "rules" to the given context. In the case of gentrification, ML has been used in London (Reades et al., 2019), Washington D.C., (Yoo, 2023) and Nashville, TN (Knorr, 2019). By defining the term gentrification and finding suitable and measurable parameters, each of the studies have their own take on how ML can be put to use. In Knorr (2019) it is emphasized how the different definitions of gentrification are displayed in the weights and values that the predictor variables are given. It is stated that an arbitrary selection cannot be avoided since the designation of gentrification does not account for different views of e.g household income being rated higher than unemployment (Knorr, 2019, pp. 17–19).

This results in a skewed view when comparing different prediction models. For instance, Knorr (2019) compares eligibility of gentrification by the use 50th percentile of median household income (relative to metro-wide areas) with 70th percentile average household income (relative to metro-wide areas) and 20th percentile average household income (relative to urban areas) (Knorr, 2019, pp. 20–21). In Reads et al. (2019) the predictor variables are each given a value and weight according to the impact based on literature.

The selected variables (household income, property value, occupational share and qualifications) are weighted as a whole to give a standard coefficient for socioeconomic status (Reades et al., 2019, pp. 928–930). Yoo (2023) uses data extracted from American Housing Survey, where the main denominator is if the current resident has a higher income than the previous one (Yoo, 2023, pp. 3–6).

While gentrification as a phenomenon can be measured using several types of accessible data, the mentioned prediction models also has blindspots. Where the use of quantitative data stands out in creating reliable overviews and visualizing patterns, it also lacks in depth and nuances. According to García (2025) in the study called *When the Map Does Not Tell the Whole Story*, the use of qualitative data in the form of interviews is central to shining light on the unmeasurable parameters concerning gentrification. The study underlines how gentrification can be tracked prior to the patterns of economic and demographic factors, and that residents in vulnerable areas often feel and articulate the change in their neighborhood beforehand (García, 2025, p. 16). The study focuses on interviewing the residents of different parts of Salt Lake City, Utah and comparing their

answers about gentrification and neighborhood change to a prediction model made for the same areas. It also showed how even the advanced prediction model does not have the ability to account for warnings of increased rent or evictions or how the business community changed in composition (García, 2025, pp. 17–19). This acknowledgment is important to have in mind when constructing a prediction model. Gentrification and its impact can reach deeper than the data shows. The different forms of prediction models also make it evident, that the weights and values of the prediction model largely depend on the extent of the data and how the literature emphasizes the prediction parameters.

3.4.1 The Use of Machine Learning in Prediction Models

As mentioned earlier there are several takes on how ML can be used to predict gentrification. This is evident in the different uses of ML models. The models are differentiated between supervised or unsupervised learning, depending on whether the pattern recognition process relies on a known ground truth. It is also essential to consider the desired outcome and the size and structure of the data used. In relation to gentrification studies, the supervised models are demonstrated in using ground truth data based on areas predetermined as gentrified. The majority of ML and gentrification studies use a supervised tree-based model to predict where future gentrification will occur (Knorr, 2019; Thackway et al., 2023; Yoo, 2023).

In Yoo (2023) & Knorr (2019) the authors test multiple models, which both found, that the Random Forest (RF) Classifier has the strongest accuracy and recall for predicting gentrification. The power of tree-based learning comes from the large amount of features in the dataset used for gentrification studies (Thackway et al., 2023). These studies consist of both a large number of observations and a wide number of features. Reades et al. (2019) was the first to apply ML into urban changes and was used to predict areas in London that were likely to experience uplift or decline by 2021. The RF model was chosen as it can quickly assign observations into classes and predict values based on observations. Furthermore, the bias of the model and the amount of tuning in RF is limited. It is both a robust and versatile model, but the versatility can be a drawback (Reades et al., 2019).

In the ML based gentrification study Thackway et al. (2023) both RF, Gradient Boosting Machine (GBM) and Extreme Gradient Boosting (XGBoost) is tuned and tested. All three models are tree-based and supervised. RF differ from the boosting methods GBM and XGBoost in the way that RF builds many decision trees independently and boosting methods builds trees sequentially, learning from the previous trees (Thackway et al., 2023, p. 3). In the study the GBM achieved the highest accuracy.

It is then argued, that Gradient Boosting Models are superior to RF due to their ability to boost accuracy from consecutive training. However the computation time can be longer for GBM than for RF (Thackway et al., 2023).

3.5 Consequences of Gentrification

In Larsen and Hansen (2008), the authors point to politicians and authorities who have a predominantly negative discourse on the marginalized inhabitants of Vesterbro in the 1990's. The local inhabitants were not prioritized, and the authorities would rather have more inhabitants from the economically sustainable population (Larsen & Hansen, 2008, p. 2433), (Hansen et al., 2001, p. 862).

On the other hand, it is true that with gentrification comes a higher living standard for the people currently living in the neighborhood (Vidal, 2019). This reflects in multiple different factors, such as education, income and housing prices. In Drew (2018) the author states that for some neighborhoods where vacancy rates are high it can have a positive effect on the diversity if the middle class moves into the vacant buildings.

By not having a clear definition of gentrification it can be derived that there may be pros and cons to the phenomenon. In the work *The death and life of great American cities* the planning theorist Jane Jacobs argues, that having an increase of wealth in a previously poor neighborhood does not in itself result in an increase nor a decrease in quality and opportunity. Jacobs states, that it is the lack of diversity in a neighborhood that compromises the living conditions (Jacobs, 1961, p. 251). Jacobs elaborates on the concept of 'the self-destruction of diversity' by explaining successful and vital city areas saying:

“(...) so many people want to make use of such places, so many people work in them or live in them or visit in them, that municipal self-destruction ensues. In killing successful diversity combinations with money, we are employing perhaps our nearest equivalent to killing with kindness. In short, the demand for lively and diversified city areas is too great for the supply” (Jacobs, 1961, p. 255).

This can be linked to high esteem that Copenhagen and its central districts is being held in presently. As mentioned in chapter 1 and section 3.2 the rising popularity in these areas will either directly or indirectly cause a transformation that can accommodate the needs of the resourceful. The paper *Aesthetics of Gentrification* (Hock et al., 2021), which maps

out the urban development in the big cities, elaborates on Jane Jacobs' idea of diversity self-destruction and how it is linked to gentrification in the modern age. In the paper it is noted, that Jacobs definition in 1961 exemplifies 'the golden age of gentrification'. She states that it was a slow movement of diversifying the underprivileged areas of the city with newcomers that had resources and will to restore. Whereas the modern gentrification as Jacobs said: "(...) explodes into a feeding frenzy of real-estate speculation and evictions", which ultimately leads to a homogenization of inhabitants (Hock et al., 2021, p. 33). This reasoning from Jacobs is what inspired the planner Michael Mehaffy to create the Jacobs Curve as a visualization of Jacobs' thoughts, which labels gentrification as a spectrum between wealth and diversity seen on figure 3.3 (Mehaffy, 2020).

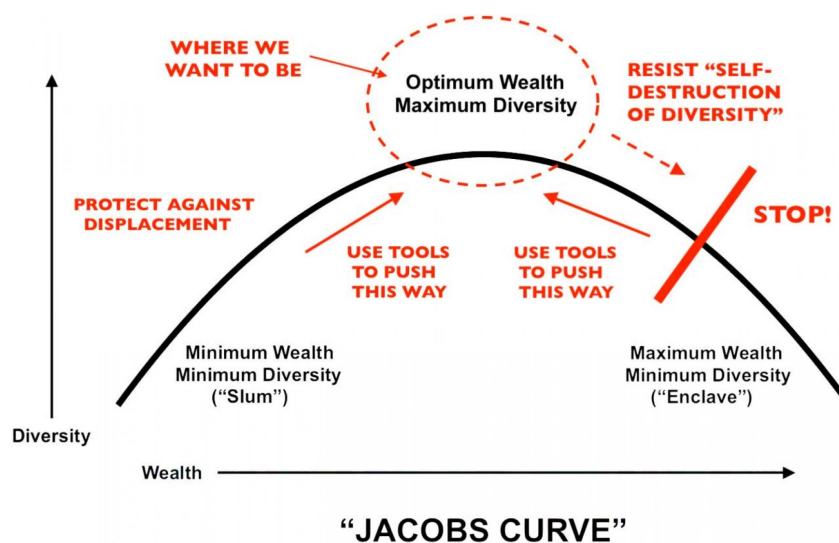


Figure 3.3: Jacobs Curve by Michael Mehaffy (Mehaffy, 2020)

As the figure shows there exists a point of optimum wealth and maximum diversity that can be achieved. Minimum diversity both exist in the gentrified 'enclaves' and the ungentrified 'slums', but Mehaffy states, that by having a coexisting of poor and rich it contributes to vibrant cities. Contradictory, persistent gentrification will with time displace the locals and thus the authenticity, that might have attracted the 'gentrifiers' in the first place (Mehaffy, 2020).

3.6 Summary of Literature Review

This literature review examines gentrification with a focus on defining the concept in both an international and Danish context and to identify the different methods for measuring and predicting gentrification. It is interpreted that there is no unique definition of gentrification. Some literature adopt a more positive or neutral definition that emphasize improvement and upgrading. Others acknowledge displacement and loss of cultural character as a direct consequence of gentrification. This aspect is seen as essential in order to get a holistic view on gentrification. This will be accounted for when the specific data is being determined to get the most accurate measure in the prediction model.

The review focuses on the drivers of gentrification and it is found that urban renewal policies and commodification of spaces in the city leads to gentrification in Copenhagen. Furthermore the findings from the review lead to a better understanding of how gentrification in Copenhagen have unfolded and what consequences the displacement of lower-income and marginalized groups have experienced. The literature points to Inner Vesterbro being the area with the clearest signs on gentrification in Copenhagen, but also Nørrebro and Amager West exhibit signs on gentrification.

In Copenhagen the consequences of gentrification consists mostly of the rent for some areas to have risen with up to 50% and the housing type changed from private renting to private housing cooperatives which lead the establishment to profit from the new change in housing types. The amount of information on gentrification in Denmark is limited, and therefore a need to expand the concept and understanding in Denmark. In addition, there are international attempts to develop machine learning models that can predict gentrified areas, but none for Denmark.

METHODOLOGY